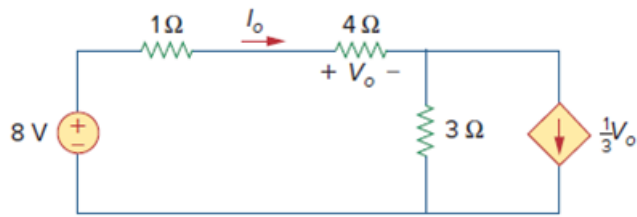


Problem

Use source transformation to find I_o in Fig. 4.96.

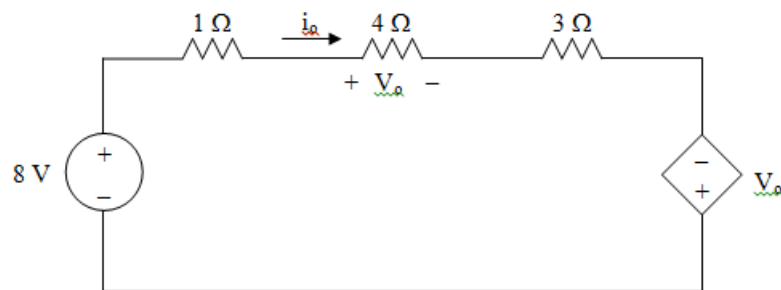
Figure 4.96:



Step-by-step solution

Step 1 of 1

Convert the dependent current source to a dependent voltage source as shown below.



Applying KVL,

$$-8 + I_o(1 + 4 + 3) - V_o = 0$$

But $V_o = 4I_o$

$$-8 + 8I_o - 4I_o = 0 \quad \longrightarrow \quad I_o = \underline{2\text{ A}}$$